

Problem 1 : Subadditive Sequences

Let $(a_n)_{n=1}^{\infty}$ be a sequence of positive real numbers such that

$$a_{n+m} \leq a_n + a_m, \quad m, n \geq 1.$$

Show that $\lim_{n \rightarrow \infty} \frac{a_n}{n}$ exists by showing

$$\lim_{n \rightarrow \infty} \frac{a_n}{n} = \inf_{n \geq 1} \frac{a_n}{n}.$$

Hint. Treat separately $\liminf$ and $\limsup$.

Solution.

$$\liminf_{n \rightarrow \infty} \frac{a_n}{n} = \lim_{n \rightarrow \infty} \inf_{k \geq n} \frac{a_k}{k} \\ =$$

Problem 2 : Equicontinuity and Uniform Convergence

Let $f_n : [0, 1] \rightarrow \mathbb{C}$ be a sequence of continuous functions. Suppose f_n converge pointwise. Show that the sequence converges uniformly if and only if the collection of functions $\{f_n\}$ is equicontinuous.

Solution. $\Rightarrow$: Suppose $f_n \rightarrow f$ uniformly. Then there exists $N \in \mathbb{N}$ such that $\forall n \geq N$ we have that for all $x \in [0, 1]$:

$$|f_n(x) - f(x)| < \frac{\varepsilon}{3}.$$

There exists a δ_f such that $|f(x) - f(y)| < \frac{\varepsilon}{3}$ for $|y - x| < \delta_f$

Since f_n are continuous functions on compact sets they are uniformly continuous. There exists δ_i such that for all $x, y \in [0, 1]$ where $|x - y| < \delta_i$: $|f_i(y) - f_i(x)| < \varepsilon$. Let

$$\delta = \min \left(\min_{i \in \{1, \dots, N-1\}} \delta_i, \delta_f \right).$$

Then for all $n \geq N$ and $x, y \in [0, 1]$ where $|x - y| < \delta$:

$$|f_n(x) - f_n(y)| \leq |f_n(x) - f(x)| + |f(x) - f(y)| + |f(y) - f_n(y)| < \varepsilon$$

Thus for δ , $\{f_n\}$ is equicontinuous.

$\Leftarrow$: Suppose $f_n \rightarrow f$ pointwise and $\{f_n\}$ is equicontinuous. Thus $\forall \varepsilon > 0$ there exists a $\delta > 0$ such that for all $n \in \mathbb{N}$ and $|x - y| < \delta$: $|f(y) - f(x)| < \varepsilon$.

Problem 3 : Integral Operator on $C([0, 1])$

Fix $g \in C([0, 1] \times [0, 1])$. For a Riemann integrable function $f : [0, 1] \rightarrow \mathbb{R}$, define the operator T via

$$(Tf)(x) = \int_0^1 g(x, y) f(y) \, dy.$$

- (a) Prove that the function Tf is continuous on $[0, 1]$.
- (b) Show that the set

$$S = \left\{ Tf; f \text{ is Riemann integrable and } \sup_{x \in [0, 1]} |f(x)| \leq 1 \right\}$$

is precompact (i.e. its closure is compact) in the space $C([0, 1])$ equipped with the uniform metric

$$d_\infty(f_1, f_2) = \sup_{x \in [0, 1]} |f_1(x) - f_2(x)|.$$

Solution. (a): Since $g \in C([0, 1]^2)$. $\forall \varepsilon > 0$, there exists a $\delta > 0$ such that $\forall x, y \in [0, 1]$ where $d(x, y) < \delta$:

$$\begin{aligned} |g(x, z) - g(y, z)| &< \varepsilon \left(\int_0^1 |f(z)| \, dz \right)^{-1} \\ |(Tf)(x) - (Tf)(y)| &= \left| \int_0^1 g(x, z) f(z) \, dz - \int_0^1 g(y, z) f(z) \, dz \right| \\ &= \left| \int_0^1 [g(x, z) - g(y, z)] f(z) \, dz \right| \\ &\leq \int_0^1 |g(x, z) - g(y, z)| |f(z)| \, dz \\ &< \varepsilon \end{aligned}$$

(b): Let (Tf_n) be a sequence in S . Note g is a continuous function on a compact set hence is bounded thus Tf also bounded in S because f is bounded. Thus (Tf_n) is bounded. Then we want to show that (Tf_n) is equicontinuous. Then by Ascoli, (Tf_n) has a uniformly convergent subsequence hence A is compact by sequential compactness.

So we need to show that $\{Tf_n\}$ is equicontinuous. We know that Tf_n is uniformly continuous for all n . By problem 2, the collection is equicontinuous

Problem 4 : Alternating Series of Continuous Functions

Let $f_n : [-1, 1] \rightarrow [0, 1]$ be continuous functions. Assume that for each $x \in [-1, 1]$, the sequence $(f_n(x))_{n \geq 1}$ is decreasing and

$$\lim_{n \rightarrow \infty} f_n(x) = 0.$$

For $n \geq 1$ and $x \in [-1, 1]$, let

$$g_n(x) = \sum_{m=1}^n (-1)^m f_m(x).$$

Show that $g_n(x)$ converges to some $g(x) \in \mathbb{R}$ for all $x \in [-1, 1]$, and that the function $g : [-1, 1] \rightarrow \mathbb{R}$ thus defined is continuous on $[-1, 1]$.

Solution.